



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CSA0311 – Data Structures

Online Examination and Auto-Evaluation Platform

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INTRODUCTION

- Digital platform to conduct online examinations
- Automatically evaluates student answers
- Reduces manual work and human errors
- Provides instant results and performance analysis
- Suitable for schools, colleges, and training institutes

ABSTRACT

- Online examinations require efficient management of questions, evaluation, and result analysis.
- This project develops an Online Examination and Auto-Evaluation Platform using the C programming language.
- Module 1 handles question bank management, including adding, storing, displaying, and searching questions
- Module 2 performs automatic answer evaluation by comparing student responses with the correct answers
- Module 3 focuses on result analysis and performance evaluation, where student scores are sorted and ranked.
- Fundamental data structures such as arrays, structures, and sorting algorithms are used.
- This project provides a simple, reliable, and efficient solution for online examination management.

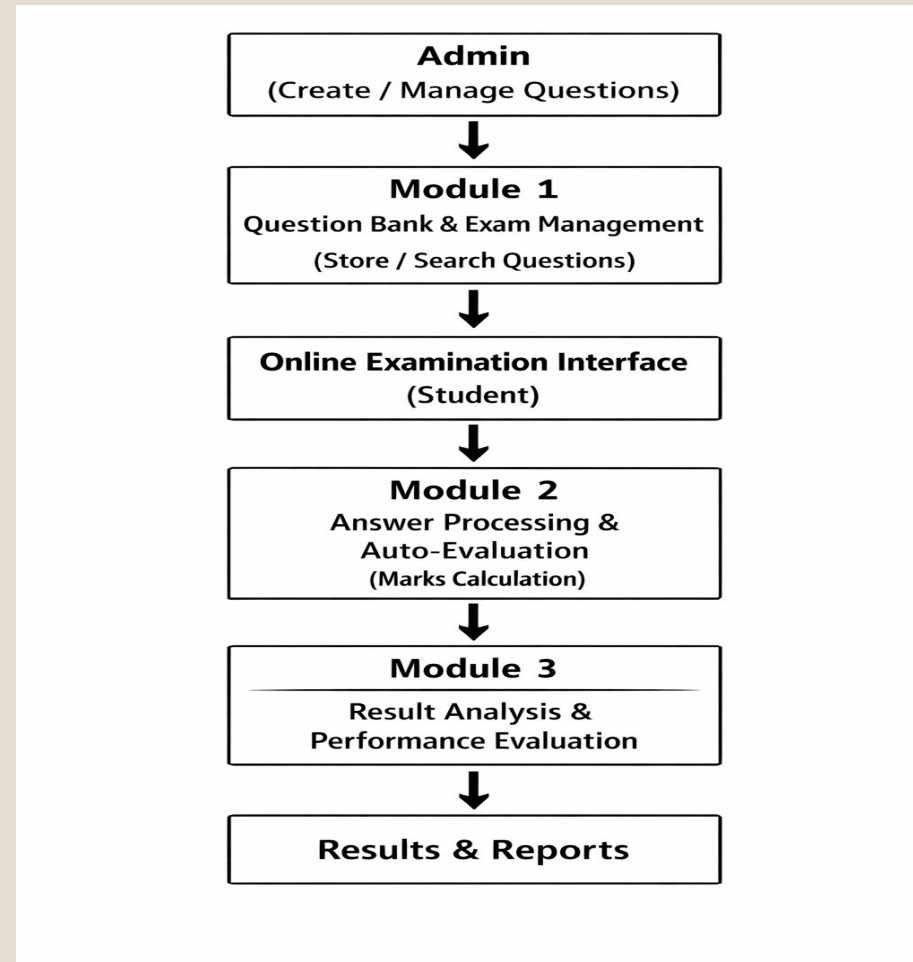
OBJECTIVE

- To design and implement an Online Examination and
- Auto-Evaluation Platform
- To apply **Data Structure concepts** for efficient storage and retrieval of exam data
- To automate the evaluation process and generate results accurately
- To reduce time, manual effort, and human errors in examination systems
- To ensure scalability, security, and efficient performance of the platform

LITERATURE REVIEW

SNO	AUTHOR NAME	YEAR	TITLE/ WORK	ALGORITHM/TE CHNIQUE USED	FINDINGS/ REMARKS
1	R. Sharma et al.	2019	Online Examination System	Arrays, Linear Search	reduced manual efforts,
2	S. Kumar	2020	Automated Exam Evaluaton System	Arrays, Comparison Algorithm	Faster and accurate result processing
3	A. Patel	2021	Web-Based Examination Platform	Queue, Stack	Improved accessibility and exam management
4	M. Verma	2022	Result Analysis System	Sorting Algorithms	Effective performance analysis using ranking

ARCHITECTURE DIAGRAM



PROBLEM STATEMENT

- ❖ Manual examination systems are time-consuming
- ❖ High chances of human errors in evaluation
- ❖ Delay in result processing and ranking
- ❖ Difficulty in managing large question banks
- ❖ Lack of instant performance analysis

METHODOLOGY

- **Question Storage:** Use structures and linked lists to store question data
- **Answer Processing:** Store student responses using stack
- **Auto-Evaluation:** Compare answers using recursion logic
- **Result Processing:** Sort scores using bubble sort for ranking
- **Performance Analysis:** Analyze time and space complexity

MODULES

Module 1: Question Bank Management

Module 2: Exam Execution & Evaluation

Module 3: Result Analysis & Ranking

MODULE 1

QUESTION BANK MANAGEMENT

- Stores questions using array of objects.
- Displays questions dynamically using loop.
- Uses DOM manipulation to show exam.
- Structured format: text, options, correct answer.
- Easy to add or remove questions.
- Supports dynamic question generation.

```
// Question Bank
let questions = [
  { text:"Capital of India?",
    options:["Mumbai","Delhi","Chennai","Kolkata"],
    correct:"B" }
];

// Load Questions
function loadExam() {
  let html = "";
  questions.forEach((q,i) => {
    html += `${q.text}</b><br>`;
  });

  document.getElementById("examArea").innerHT
  ML = html;
}
```

MODULE 2

EXAM EXECUTION & EVALUATION

- Uses recursion to calculate marks.
- Checks one question at a time.
- Adds 1 mark for correct answer.
- Stops when all questions evaluated.
- Validates answers before submission.
- Stores student score for ranking.

```
function evaluateRecursive(i) {  
  if(i >= questions.length)  
    return 0;  
  
  let mark = (stack[i] ===  
    questions[i].correct) ? 1 : 0;  
  return mark +  
    evaluateRecursive(i+1);  
}
```

MODULE 3

RESULT ANALYSIS & RANKING

- Uses Bubble Sort algorithm.
- Compares adjacent scores.
- Swaps to arrange in descending order.
- Highest score gets Rank 1.
- Suitable for small number of students.
- Displays ranking without page refresh.

```
function bubbleSort(arr){  
  for(let i=0;i<arr.length-1;i++){  
    for(let j=0;j<arr.length-i-1;j++){  
      if(arr[j].score < arr[j+1].score){  
        let temp = arr[j];  
        arr[j] = arr[j+1];  
        arr[j+1] = temp;  
      }  
    }  
  }  
}
```

RESULT

- Students receive **instant scores** after exam submission
- Performance is displayed **topic-wise and overall**
- Rankings are automatically calculated using **sorting algorithms**
- Detailed analytics help identify **strengths and weaknesses**
- Reduces **manual effort and errors**, ensuring faster feedback

SDG-SUSTAINABLE DEVELOPMENT GOAL

- Provides a **digital online examination system**
- Ensures **automatic and error-free evaluation**
- Delivers **instant results and performance analysis**
- Reduces manual work and saves time
- Promotes **technology-based smart education**

CONCLUSION

- Provides efficient and automated exam management
- Ensures fast and accurate evaluation
- Uses data structures for optimized performance
- Generates instant results and rankings
- Scalable and reliable solution for institutions

REFERENCES

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THANK YOU